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Concentration determination of inorganic acids that do not absorb near-infrared (NIR) radiation through recognizing perturbed NIR water bands by them and investigation of accuracy dependency on their acidities

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ABSTRACT

When near-infrared (NIR) spectroscopy is employed to determine the concentrations of inorganic acids in etchant solutions, the perturbation of the water bands resulting from the interactions with these acids is the only information available for analysis, since inorganic acids do not directly absorb NIR radiation. The water bands perturbed by dissociated hydronium ions and respective anions of the acids are not highly component-specific; therefore, characterization of their different interactions with water is necessary to understand the results of the multivariate quantitative analysis. For this purpose, the variations in the water bands that occurred with the inclusion of HCl, H_2SO_4, H_3PO_4, and HNO_3 were initially examined with the aid of a two-dimensional (2D) correlation analysis. According to this analysis, components with higher acidity, such as HCl, perturbed the water hydrogen bonding network more strongly. Also, the interactions between each dissociated ionic species and water molecules were somewhat dissimilar. This dissimilarity enabled differentiation between the four acids, although the spectral specificity was not high owing to the absence of direct NIR absorption. The accuracy in the determination of concentrations was better for components with lower pK_as, such as HCl and H_2SO_4, owing to their stronger perturbation of the water bands. When multicomponent samples were measured, the resulting accuracies deteriorated due to the further degradation in component specificity. Acquisition of reproducible spectra and avoidance of model over-fitting are critical for reliable NIR analysis of these types of samples, as the analyte-induced spectral features are broad and rather specificity-insufficient.

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1. Introduction

In the use of near-infrared (NIR) spectroscopy for the determination of component concentrations in aqueous samples, water acts as an interferant that strongly attenuates NIR radiation, degrading the signal-to-noise ratio of a collected spectrum and the accuracy eventually. However, water can be a beneficial medium for the measurement of analytes that do not directly absorb NIR radiation, such as ionic species, as the altered spectral features of water that result from interaction with these analytes are potentially valuable sources for enabling the intended analysis [1–6]. Wafer etchant solutions are composed of several inorganic acids, and the determination and control of their concentrations are necessary to achieve optimal wafer cleaning performance. Inorganic acids are strongly dissociated into hydronium ions (H_3O^+) and corresponding anions in aqueous solution, and these dissociated ionic species do not generally directly absorb NIR radiation. Therefore, water bands perturbed by these species are the only information available for use in determination of their concentrations. As a result, NIR spectroscopy has been investigated as a fast and non-destructive analytical method for measuring the concentrations of components in diverse etchant solutions [7–10].

The NIR bands of water are broad, and the interactions of dissociated ionic species with water molecules are not highly structure-specific. Therefore, the spectral features of perturbed water bands are not distinct. In particular, when more than one acid is present in a sample, it is less easy to discern individual components in the acquired spectrum. As a result, the use of multivariate analysis is required to differentiate the components and determine individual concentrations. When using specificity-deficient spectra for modeling, characterization of the varied

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water spectral features with a variety of inorganic acids and subsequent examination of their differences is necessary to understand the result of multivariate analysis. However, these studies have been rarely reported to date.

This study aims to investigate NIR water bands perturbed by the inclusion of four different inorganic acids (HCl, H_2SO_4, H_3PO_4, and HNO_3) and to examine their dissimilar characteristics with the aid of two-dimensional (2D) correlation analysis [11]. The 2D correlation analysis was employed to elucidate unseen spectral variations, as the perturbed water bands were expected to be broad with only minute component-specific features. The accuracy of the concentration determination for each acid in aqueous solution was also acquired, and the variation in these accuracies was evaluated in relation to the corresponding acidities (pK_a). The acidity could be related to the accuracy because it governs the strength of the interaction between the dissociated species and water molecules, and thus the corresponding degree of perturbation of the water bands. Binary component samples were also used to evaluate the accuracy for determination of individual acid concentrations. It was reasonable to expect that the resulting accuracy would decrease, as the spectral selectivity with two components becomes worse. Finally, requirements for ensuring reliable NIR spectroscopic analysis of these types of samples were discussed.

2. Experimental methods

High purity HCl, H_2SO_4, HNO_3, and H_3PO_4 were obtained from a common supplier. For each case, twenty sample solutions with concentrations ranging from 0.1 to 2.0 M (increment: 0.1 M) were prepared, and NIR spectra (resolution: 8 cm^{-1}) of the samples were collected using an ABB FT-NIR spectrometer (Quebec, Canada) equipped with a tungsten–halogen source and an InGaAs detector. Each sample was placed into a glass vial (inner diameter: 9.6 mm) and then into a metal vial holder with a temperature control precision of ±0.1 °C. The temperature of all samples was maintained at 25.0 °C during collection of spectra. An empty glass vial was used as a background. Triplicate spectra were collected for each sample, so a total of sixty spectra was collected from the twenty samples in each case. All spectral preprocessing and multivariate analyses, including baseline corrections, Savitzky–Golay smoothing, generation of second derivative spectra, 2D correlation analysis, and partial least squares (PLS), were performed using MATLAB R2016a (Math Works Inc., MA, USA).

3. Results and discussion

3.1. NIR spectral features of inorganic acids

The baselines of the sample spectra varied slightly as a result of the use of round disposable glass vials for simple spectral acquisition. These type of vials do not have perfectly consistent optical quality or surface curvature. As the magnitude of the variation in the water band for each acid was expected to be small, the baselines of spectra were corrected to a value of zero at 9200 cm^{-1} to allow for the examination of spectral features. Fig. 1 shows the baseline-corrected NIR spectra (range: 9200–7900 cm^{-1}) for HCl, H_2SO_4, HNO_3 and H_3PO_4 samples at varying concentrations from 0.1–2.0 M. For each acid, the 0.1 and 2.0 M samples are shown in blue and red colors, respectively, while the grey spectra correspond to the remaining samples. The pure water spectrum is shown in black (arbitrarily offset for clarity) alongside the sample spectra for comparison. The broad bands in this range correspond to the overtones of the fundamental OH vibrations, as previously reported [8,9]. For all four acids, the most notable feature is the decrease in the absorbance of the water bands with increasing acid concentration. This is largely due to the displacement of water molecules by the inclusion of the acid. Comparing the spectra of the 2.0 M samples of each acid suggests that the shapes of the water bands are broadly similar. However, their detailed features are minutely different, indicating somewhat dissimilar interactions between the dissociated species of each acid and the water. It is worthwhile to reiterate here that the

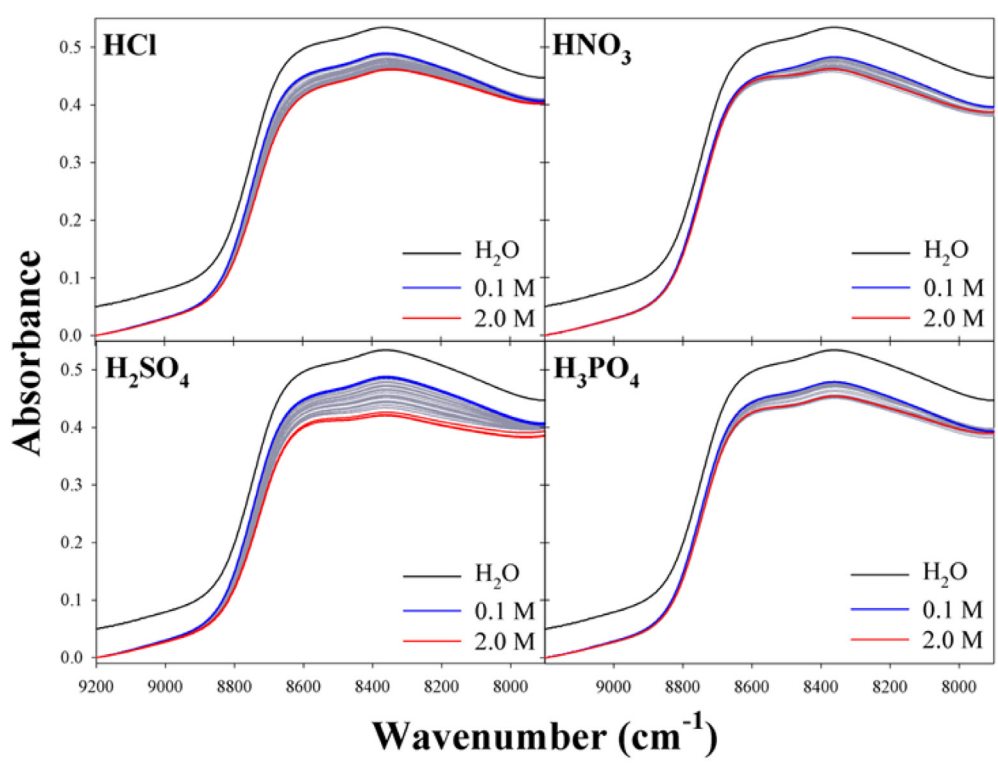

Fig. 1. Baseline-corrected NIR spectra (9200–7900 cm^{-1}) of HCl, H_2SO_4, HNO_3 and H_3PO_4 samples at varying concentrations from 0.1–2.0 M. In each case, the 0.1 and 2.0 M sample spectra are shown in blue and red, respectively, while the grey spectra correspond to the remaining samples. The pure water spectra (black, arbitrarily offset for clarity) are also shown for comparison. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

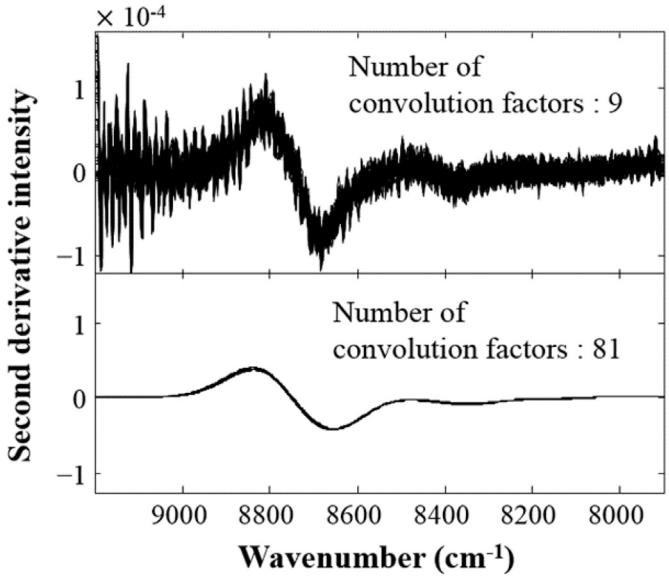

Fig. 2. Second derivative spectra obtained from the raw NIR spectra of all HCl samples (Fig. 1(a)) using 9 (top) and 81 (bottom) convolution factors.

dissociated species of these acids (H_3O^+ and corresponding anions) absorb NIR radiation in this spectral range either negligibly or not at all.

Because the features of the varied water bands by inclusion of the acids were broad and the component-specificity was not clear, the examination of detail spectral features was necessary. For this purpose, a derivative based on the Savitzky–Golay method was employed [12]. The Savitzky–Golay method arose from the idea that variables in a particular vicinity of a spectrum could be fit with low-degree polynomials. In its application, when the number of convolution factors is increased to reduce noise, such as in a second derivative spectrum, the final peak shape is distorted, adversely affecting the accuracy of the spectroscopic analysis. Fig. 2 displays the second derivative spectra obtained from the raw NIR spectra of the HCl samples in Fig. 1, using 9 (top) and 81 (bottom) convolution factors. With 9 convolution factors, the second derivative spectrum exhibits significant choppiness. The use of a smaller number of convolution factors clearly amplifies the underlying noise. In contrast, when the number of convolution factors is substantially larger, such as in the spectra employing 81 convolution factors, the noise nearly disappears and the baselines of the spectra become consistent. However, the magnitudes of the peaks decrease, and the peak shapes are somewhat broadened. The decreased intensity and broadened peaks negatively affect the accuracy of the concentration determination, as the acid-induced spectral variations are already indistinct. Therefore, the number of convolution factors needs to be optimally chosen to maximize the acid-induced features with minimal noise in the final spectra before determining the concentrations using partial least squares (PLS).

3.2. Determination of acid concentration in single-component samples

In addition to the number of convolution factors used to generate the second derivative spectra, the number of principal components (PCs) used in the PLS model needs to be properly determined to avoid under- or over-fitting. Therefore, an optimal combination of the numbers of convolution factors and PCs was evaluated with a grid search [13]. Prior to the grid search, two thirds ($n = 40$) of the spectra were assigned to a calibration set, while the remaining spectra ($n = 20$) were assigned to a prediction set. The calibration spectra were then subjected to a grid search. Fig. 3(a) shows the contour map illustrating the root mean square error of cross validations (RMSECVs) that occur with simultaneously changing of the numbers of both convolution factors (from 1 to 80) and PCs (from 1 to 10) for the determination of HCl concentrations. The deeper blue color indicates lower RMSECV values. Examining the color change along the *y*-axis, the RMSECV decreases with increasing number of convolution factors and, for a given number of PCs, no significant variation is observed after a certain point. This indicates that the use of additional convolution factors beyond this point does not help to elucidate further analyte-induced spectral features for PLS modeling. In addition, the color change along the *x*-axis informs that the decrease in the error is insignificant for the use of more than three PCs.

Based on the contour plot assessment, the use of second derivative spectra generated with 45 convolution factors and three PCs in the PLS modeling (indicated by the white circle in the plot) provided the most accurate determination of the HCl concentrations (root mean square error of prediction (RMSEP): 0.010 M), as summarized in Table 1. It is worth noting that three PCs were used in the model, even though only HCl was present in the samples. If the displacement of water molecules by the inclusion of HCl was the only phenomenon occurring in the solutions, then use of a single PC would be sufficient to follow the concentration variation. The need for two additional PCs in the modeling implies that the interactions of two dissociated species (H^+ and Cl^-) with water molecules exist and are likely multi-modal.

Fig. 3(b) shows the second derivative spectra of the HCl samples generated using 45 convolution factors. The blue and red colored spectra again correspond to the 0.1 and 2.0 M HCl samples, respectively. As shown, the intensities of the OH bands around 8680 and 8380 cm^{-1}

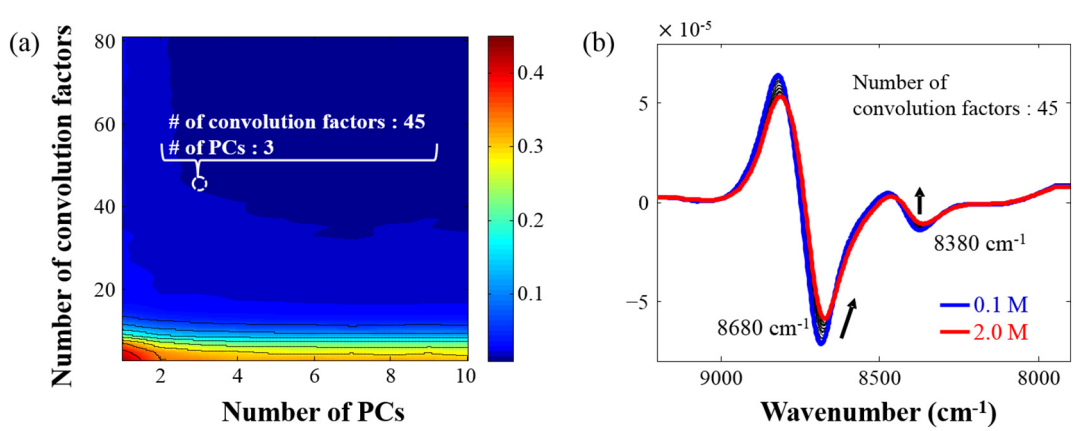

Fig. 3. (a) Contour map showing the resulting RMSECVs when the numbers of convolution factors (1–80) and PCs (1−10) are changed simultaneously, and (b) second derivative spectra of HCl samples generated using 45 convolution factors. The blue and red colored spectra correspond to the 0.1 and 2.0 M HCl samples, respectively. The white circle indicates the optimal number of convolution factors and PCs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
PLS results for the determination of acid concentrations in single-component samples. The number convolution factors and PCs in each case are also given. The numbers in parentheses are R^2 values of the corresponding concentration correlation plots (refer to Fig. S1).

	Number of convolution factors	Number of PCs	RMSECV (M)	RMSEP (M)
HCl	45	3	0.010 (0.9997)	0.010 (0.9997)
H_2SO_4	45	2	0.012 (0.9996)	0.015 (0.9996)
HNO_3	29	3	0.023 (0.9984)	0.024 (0.9982)
H_3PO_4	37	3	0.044 (0.9947)	0.044 (0.9954)

decrease with increasing HCl concentration. This indicates that the displacement of water molecules becomes greater with increasing HCl concentration [5,14–16]. It is also notable that the positions of both peaks gradually shift toward higher wavenumbers with increasing HCl concentration. This indicates the existence of interactions between the dissociated species and the water hydrogen bonding network. These observed variations in the HCl spectra support the necessity of using multiple PCs in the PLS modeling.

Since the dissociated H^+ and Cl^- ions do not absorb NIR radiation, it is worthwhile to examine which spectral information the model utilizes to assess the varying HCl concentration. For this purpose, the scores of each sample and corresponding loadings were examined, and are shown in Fig. 4(a) and (b), respectively. The variation in the first scores is linear with the change in HCl concentration, and the first loading predominantly weights on the main water bands located around 8725 and 8380 cm^{-1}. This suggests that the first PC describes the displacement of water caused by the inclusion of HCl, which is a distinct source for enabling determination of the HCl concentrations.

Since the dissociated species obviously interact with water molecules, the subsequent PCs are necessary to describe the corresponding spectral variations to achieve more accurate concentration determinations. The patterns in the concentration-dependent variations of the second and third scores are approximately opposite and vary significantly. The symmetrical nature of the trends in these scores suggests that the interaction of H^+ and Cl^- ions with water occur in opposite manners. In addition, the band positions of maximal weight in the second loading, such as that around 8650 cm^{-1}, are different from those in the first loading. Furthermore, the number of weighted bands in the third loading is increased and their locations are different compared to those in the first and second loadings. Based on these results, it is reasonable to presume that the second and third loadings explain the spectral variations that arise from the interaction of water molecules with the dissociated species.

Table 1 summarizes the RMSECVs obtained in the concentration determinations for the other inorganic acids in the single-component samples. For each case, the optimal numbers of convolution factors and PCs are presented. The concentration-dependent score variations in the measurements of H_2SO_4, HNO_3, and H_3PO_4 samples are quite similar to those for the measurement of HCl, as shown in Figs. S2–S4 in the Supplementary information. This suggests that water displacement and the interaction of dissociated species with the water hydrogen bonding network also occur with these acids. Comparing the second or third loadings between the acids, it is reasonable to conclude that the specific interaction patterns of the dissociated species of each acid differ.

One interesting observation is that the RMSEPs resulting from the concentration determinations for each acid are different, as summarized in Table 1. To probe the origin of these differences, the acid dissociation constant (K_a) of each acid was considered. Fig. 5 illustrates the relationship between the RMSEPs obtained in the individual measurements and the corresponding pK_a values of the acids. As shown, the RMSEP is clearly related to the pK_a. The error was lowest in the measurement of HCl concentrations, which has the smallest pK_a, while the accuracy was worst in the measurement of H_3PO_4 concentrations, which is the weakest of the four acids. Stronger dissociation results in stronger interactions between the dissociated species and the water; therefore, the resultant variations in the water band are more distinct and more accurate concentration determination is possible. Because the dissociated species of the acids do not absorb NIR radiation, the strength of acid dissociation is a major factor influencing the perturbation of the water band, as well as the accuracy.

3.3. Two-dimensional (2D) correlation analysis

To provide additional insight into the investigation of the sample spectra, 2D correlation analysis [11] was employed to further probe the interactions of the acids with water molecules. A negative peak in the second derivative spectra typically generates two positive peaks to its right and left sides in a 2D correlation map [17], and these positive peaks are unwanted artifacts that do not represent meaningful chemical changes and can seriously hamper the analysis of the 2D map. Therefore, the use of second derivative spectra for 2D correlation analysis is not generally recommended. Therefore, the baseline-corrected spectra in Fig. 1 were used for the 2D correlation analysis. Fig. 6(a) and (b) show the synchronous and asynchronous correlation maps,

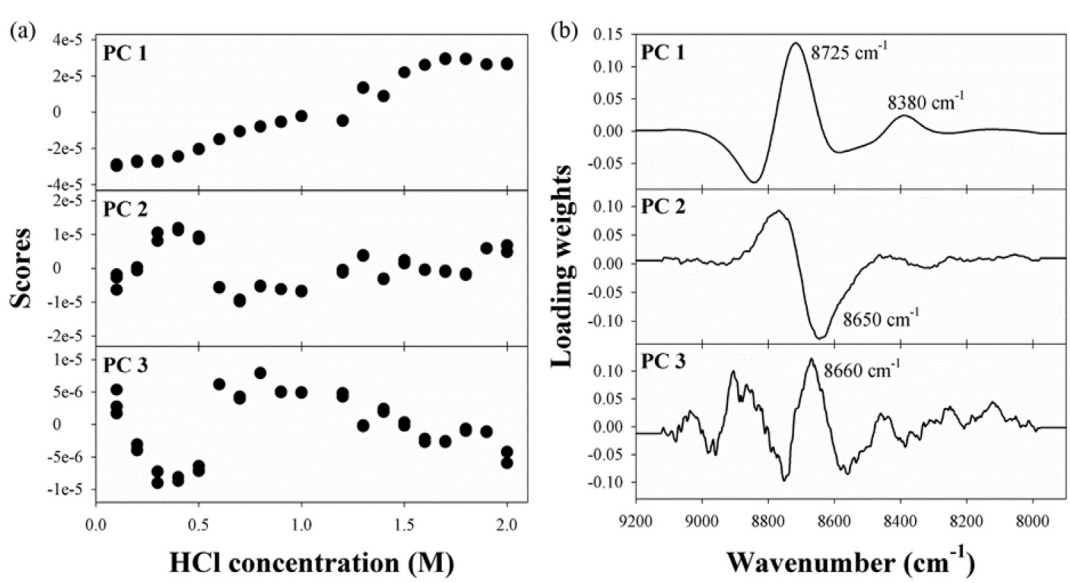

Fig. 4. (a) First three scores of each sample, and (b) corresponding loadings in the determination of HCl concentration using PLS.

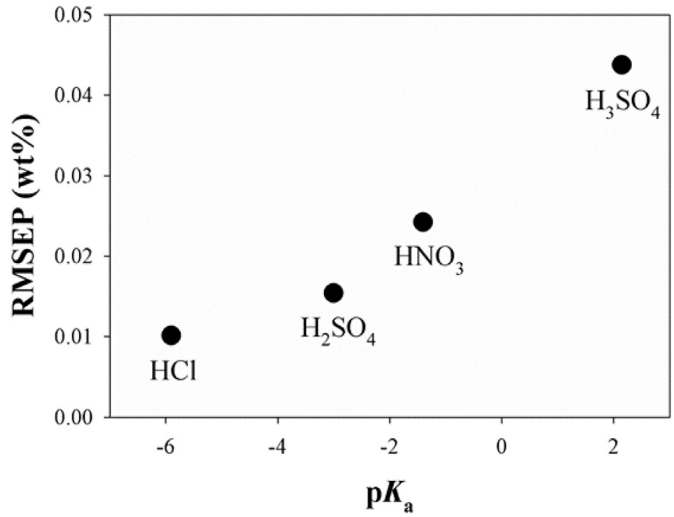

Fig. 5. Relationship between the RMSEPs in the concentration determination for each acid and the corresponding pK_a of the acid.

respectively, derived from the HCl sample spectra. Reference spectra are placed at the top and right side of the plots. Broad auto-correlation peaks around 8730 cm^{-1} are observed in the synchronous correlation map, indicating that the variations in the spectra are in the same direction, and there is collinearity among the spectral variables. This also coincides with the appearance of the dominant weight from the OH bands observed in the first loading (Fig. 4(b)). The synchronous correlation, therefore, originates from the decrease in intensity due to the displacement of water with the increasing HCl concentration.

Another notable observation is the appearance of several cross peaks in the asynchronous correlation map. The unshaded and shaded areas indicate positive and negative correlation intensities, respectively. The peaks observed at 8650 cm^{-1} and 8380 cm^{-1} show the dissimilar behavior between these bands, i.e., the variation in the 8650 cm^{-1} band occurs prior to that of the 8380 cm^{-1} band. The delay in the intensity change indicates these two OH bands have different behaviors in the HCl samples. In addition, the development of cross peaks at 8650 cm^{-1} and 8720 cm^{-1} implies that the intensity change of the 8720 cm^{-1} band occurs before that of the 8650 cm^{-1} band. This suggests that the broad peak at around 8700 cm^{-1} is composed of multiple (at least two) bands, and each of these bands has different origins. The appearance of asynchronous correlation peaks could arise from the alteration of the water hydrogen bonding network by interactions with two dissociated species (H_3O^+ and Cl^-), eventually causing a shift of the OH band to lower wavenumbers, as shown in Fig. 3(b). It is clear that the interactions of each dissociated ion with water differ oppositely, as also observed in Fig. 4(a), and therefore the PLS model requires multiple PCs to characterize the perturbed water band in different manners.

The results of 2D correlation analysis for the H_2SO_4, HNO_3, and H_3PO_4 samples showed similar correlation patterns, and had the same coordinates for local maxima and minima (refer to Figs. S5–S7 in the Supplementary information). To examine the 2D correlation results for all four acids together, the auto-correlation spectra for each acid are compared, as shown in Fig. 7. These auto-correlation spectra correspond to the intensities of the peaks along a diagonal in the synchronous correlation spectra. In other words, the peaks in the auto-correlation spectra represent the susceptibility of OH bands to the addition of the acids. The correlation intensities for HCl are strong compared to those for the other three acids, particularly for the peak at 8720 cm^{-1}. This indicates that the dissociated species of HCl perturb the hydrogen bonding network in water most significantly owing to the strongest acid dissociation (lowest pK_a). For H_2SO_4 and HNO_3, two main correlation bands with considerable intensities were observed. In both cases, the positions of the correlation bands on the left are quite similar and occur around 8720 cm^{-1}, which is slightly different from the results for HCl. In addition, broad correlation bands appear on the right side of both the H_2SO_4 and HNO_3 spectra, and their positions are fairly dissimilar. The intensity of this band is considerably lower in the HCl auto-correlation spectrum. For H_3PO_4, both correlation bands are also observable. However, their intensities are substantially lower, equivalent to the weak perturbation of water, owing to H_3PO_4 having the weakest acidity among the four acids. Two observations can be made from Fig. 7: first, the auto-correlation intensities are semi-quantitatively correlated with the pK_as of the acids; and second, the shapes as well as intensities of the correlation bands all differ from each other. Overall, the investigation confirms that the dissociated species of each acid interacts with water dissimilarly.

3.4. Determination of acid concentrations in two-component samples

Since actual etchant solutions are occasionally composed of multiple acids, it is worthwhile to investigate the variation in the accuracy of concentration determination with increased compositional complexity. Since the spectral variations induced by each acid were not distinct from one another, a degradation in accuracy was expected in this situation. For this investigation, two-component samples were prepared in a combinatorial fashion, and the resulting accuracies of concentration determination for each acid are summarized in Table 2. For each case, the optimal numbers of convolution factors and PCs are presented. The acids given in bold letters are the target components in the PLS modeling. The interpretation of the results is quite complicated; however, some trends are discernable. When H_2SO_4, HNO_3, or H_3PO_4 were the

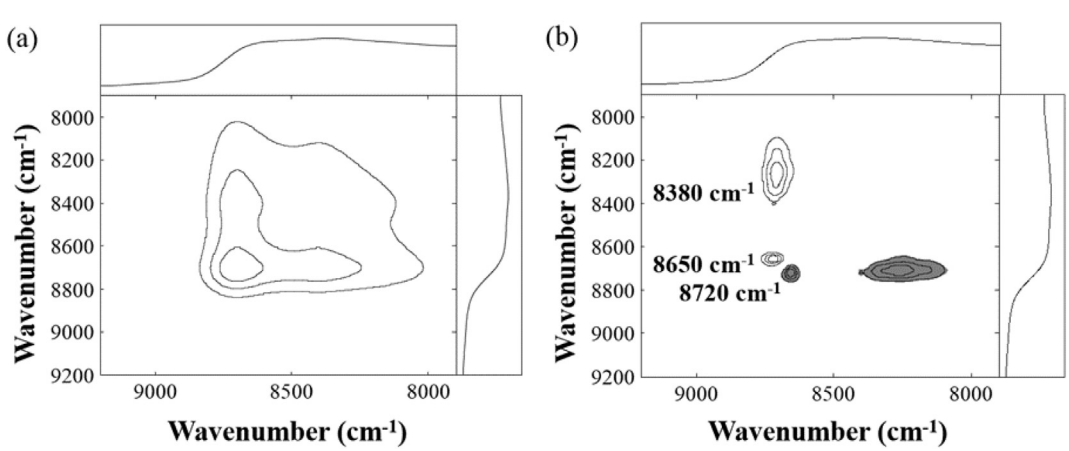

Fig. 6. (a) Synchronous and (b) asynchronous correlation maps derived from HCl sample spectra. Reference spectra are located at the top and right sides of the plots.

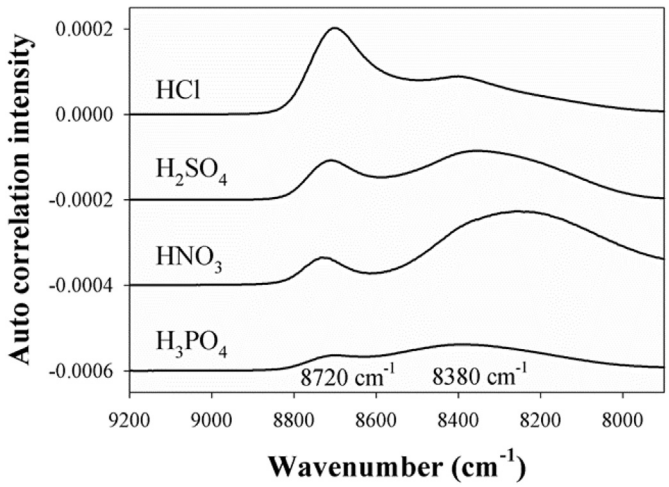

Fig. 7. Auto-correlation spectra obtained from the 2D correlation analyses of the HCl, H_2SO_4, HNO_3, and H_3PO_4 sample spectra.

target components in binary samples and the second component was HCl, the errors were smaller. Because HCl perturbation of the hydrogen bonding in water is the most distinct and strongest of all the acids, as shown in Fig. 7, the HCl-induced spectral variation is relatively easy to differentiate, resulting in an easier recognition of the other component in PLS modeling, and thus better accuracy. In addition, for the four cases resulting in a RMSEP of less than 0.050 M, three of these contain HCl either as the target or surrounding component. Furthermore, the RMSEP was lowest for the determination of HCl concentration in HCl/HNO_3 samples.

Meanwhile, the three cases with the highest RMSEPs were the determination of H_2SO_4 concentration in H_2SO_4/HNO_3 samples, HNO_3 concentration in HNO_3/H_2SO_4 samples, and H_3PO_4 concentration in H_3PO_4/H_2SO_4 samples. Based on observation of their auto-correlation spectra (Fig. 7), the patterns in the perturbation of the water bands by H_2SO_4 and HNO_3 are relatively similar, and thus the distinction between H_2SO_4 and HNO_3 is difficult. The worst accuracy is obtained in the measurement of H_3PO_4 concentration in the H_3PO_4/H_2SO_4 samples because H_3PO_4 is the weakest acid and the presence of another component (H_2SO_4) further degrades the spectral specificity. In all cases, three PCs were used for the PLS modeling, which is unchanged compared to those employed in the measurements of single-component samples. This suggests that, although two different components are present in each sample, the information available for quantification of each component does not increase, and this results from the similarities in the varied water spectral features by the individual acids. Overall, although some trends are observable, the interpretation of the PLS results for two-component samples is quite difficult owing to the insufficient component-specificity in the sample spectra. When three-component samples were measured, the measurement errors increased as expected (the relevant data are not shown in this publication).

Table 2
PLS results for the determination of acid concentrations in two-component samples. The acids given in bold type are the target components in the PLS modeling. The number of convolution factors and PCs in each case are also given.

	Number of convolution factors	Number of PCs	RMSECV (M)	RMSEP (M)
HCl/H_2SO_4	31	3	0.072	0.074
HCl/HNO_3	31	3	0.036	0.038
HCl/H_3PO_4	57	3	0.090	0.070
H_2SO_4/HCl	33	3	0.046	0.047
H_2SO_4/HNO_3	33	3	0.070	0.078
H_2SO_4/H_3PO_4	39	3	0.043	0.049
HNO_3/HCl	25	3	0.040	0.040
HNO_3/H_2SO_4	33	3	0.072	0.079
HNO_3/H_3PO_4	25	3	0.057	0.060
H_3PO_4/HCl	39	3	0.074	0.064
H_3PO_4/H_2SO_4	31	3	0.082	0.088
H_3PO_4/HNO_3	25	3	0.064	0.073

4. Conclusion

In the measurements of single-component samples, the acidity of the acid was a major factor influencing the interaction strength of the dissociated ionic species with water molecules and, in turn, the accuracy of determination of the acid concentration. The results of 2D correlation analysis indicate that a component with high acidity, such as HCl, perturbed the water hydrogen bonding network more significantly and induced more sensitive variations in the water band, leading to more accurate concentration measurements. In addition, the interactions of the dissociated species of each acid with the water were somewhat dissimilar. This enabled a differentiation between HCl, H_2SO_4, H_3PO_4, and HNO_3, although the spectral specificity among them was not high owing to the absence of direct NIR absorption features. Finally, the most critical requirement for a robust NIR analysis of samples such as etchant solutions is the acquisition of highly reproducible sample spectra, as the acid-induced spectral variations are specificity-insufficient and broad. In addition, the utilization of a method to enhance the selectivity of a multivariate model, such as pure component selectivity analysis [18], would also be beneficial for improving the reliability of NIR analysis. These issues will be investigated in future research.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.microc.2018.03.019.

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